

# THOMAS TRAN

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U.S. citizen | Open to relocation within California

## EDUCATION

### California State University, Fullerton

M.S. Computer Science (part-time)

Fullerton, CA

Expected May 2028

### University of California, Irvine

B.S. Software Engineering

Irvine, CA

March 2026

Dean's Honor List: Spring 2025, Winter 2026

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Java, C/C++, SQL

**Web & Backend:** React, Node.js, Express, Cloudflare Workers, REST APIs

**Data & Tools:** MongoDB, Firestore, PostgreSQL, Firebase Auth, Docker, Git, GitHub Actions

**Testing:** Jest, Pytest, Supertest, Firebase emulators, unit and integration testing

## EXPERIENCE

### Shoppyst | Software Engineer (Industry Capstone)

Sep 2025 - Mar 2026

UC Irvine industry partnership | Irvine, CA

- Shipped a TypeScript Chrome extension to **500+ users** with engineers and data scientists, contributing from initial development through production release.
- Reduced runtime API calls **20%** and page load time from about **1 second to under 500 ms** by precomputing MongoDB telemetry into static JSON for 15+ product configurations.
- Resolved integration race conditions by redesigning message passing and coordinating state between background service workers and content scripts.
- Built a Cloudflare Workers and MongoDB backend for click tracking, with offline write buffering and synchronization when connectivity returns.

## PROJECTS

### Vision Bucket | Full-Stack Movie Platform

React, TypeScript, Node.js, Express, Firestore | [Frontend](#) | [Backend](#)

- Built a React and Express application for movie tracking, ratings, profiles, and discussion boards; extended the original team project with backend security, tests, and CI.
- Implemented Firebase ID-token verification, role-based authorization, and resource ownership checks to restrict edits to authorized users.
- Added Supertest and Firebase-emulator integration tests in CI, a reversible Firestore migration, and OpenAPI documentation; added rate limiting and structured request logging.

### Search Engine & Web Crawler

Python, information retrieval | [Crawler source](#)

- Built ranked search over **50,000+ documents** using tokenization, stemming, and TF-IDF scoring, with average query latency below **500 ms**.
- Reduced indexing memory use **40%** using partial indexes and on-disk merging; kept the final index below **500 MB**.
- Developed a multithreaded crawler with URL canonicalization, deduplication, domain filtering, politeness controls, and restartable state to collect 50,000+ pages.

## LEADERSHIP

### Eagle Scout & Senior Patrol Leader | Troop 413

2016 - 2022

- Led a 25-person troop and coordinated 30 volunteers for an Eagle Scout service project, managing logistics, budget, and stakeholders.